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ORIGINAL RESEARCH

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Concussions in the National Football League: the evolution of video review for assessing the frequency and reliability of visible signs

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ABSTRACT

Background: The use of video review to document visible signs (VS) of sport-related concussion in the National Football League (NFL) is a novel method to recognize head injuries.

Hypothesis/Purpose: The current pilot studies used varying methodologies to (1) examine the frequency of VS in concussed NFL players using the Australian Football League's (AFL) checklist, and (2) assess the reliability of VS between non-expert and expert raters.

Study design: Cohort study

Methods: In the first pilot study, two *non-expert raters* rated VS of SRC occurring in the 2015 NFL season (n = 96) using a single VS from the AFL checklist. Based on this pilot study, two *expert raters* then rated VS of SRC during the 2017 NFL season (n = 211) using all VS from the AFL checklist. The frequency, total percent agreement (TPA), and reliability (kappa coefficients) were calculated for all VS of concussion for the two seasons. Kappa agreement was classified as fair (.41–.60), moderate (.61–.80), or substantial (.81–1.00). Significance was set at $p < .05$.

Results: The most frequent VS of concussion identified by both non-expert and expert raters were *no behavior observed, slow to get up, and motor incoordination*. The least frequent VS were *impact seizure, blank/vacant look, and facial injury*. *For non-expert raters*, the average TPA for VS ranged from 84% to 100% and kappa coefficients ranged from .52 to .68. For *expert raters*, the average TPA ranged from 83% to 100%, and kappa coefficients ranged from .56 to .86.

Conclusion: In these preliminary analyses, use of multiple VS was a superior methodology, and the reliability of VS rating was stronger for experts. Due to the inherent differences in gameplay and protective equipment used in the NFL compared to other professional sports, it is our hope these data can generate new ways to improve existing practices and identify potentially novel VS of SRC.

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Sport-related concussion; football; National Football League; video review; visible signs; traumatic brain injury; head injury

Introduction

The steps required to make an initial diagnosis of sport-related concussion (SRC) continues to be a germane topic in sports medicine. Recent expert consensus statements define a SRC as, *'a traumatic brain injury induced by biomechanical forces,'* with four noteworthy features: a blow to the head or body; a rapid and transient neurologic impairment; negative standard brain imaging; and a range of symptom endorsement [1].

Diagnosing a SRC can be difficult due to athletes not wanting to acknowledge/recognize symptoms and risk removal from play, a lack of qualified sideline medical professionals, and delayed symptom onset [2,3]. At the professional level, identifying athletes with suspected SRC can be further complicated by congested sidelines, faster game speeds, and other external factors [4,5]. Though professional athletes represent a small percentage of all sport participants, the rules and regulations of professional leagues often set the precedent for other levels of play. Alternative approaches to recognize, remove, and evaluate professional athletes with suspected SRC have been explored by several international sporting organizations. While several studies have used video to study mechanisms of injury [4], video review is currently a novel and potentially objective tool to assist in identifying a SRC during gameplay [6].

Video review allows medical professionals to examine the time immediately before and after a head impact, which may reveal key visible signs (VS) that a brain injury has occurred [7]. Video review has been used to describe the mechanisms and VS of concussion in the National Hockey League (NHL) [8–10], National Rugby League (NRL) [6,7,11–13], and Australian Rules Football (ARF) [5,14]. Other researchers have used video review to correlate VS with SRC diagnosis [15–17]. The VS recently proposed to identify a suspected SRC in professional hockey, rugby, Australian football, and American football include *loss of consciousness, lying motionless, motor incoordination, stumbling*

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gait, no protective action, impact seizure, tonic posturing, blank/vacant look, slow to get up, and *clutching at head*, among others [18]. Though attempts have been made to standardize and fully integrate video review into international sporting organization medical surveillance, its use remains exploratory.

Establishing reliability among the personnel utilized by each professional league to asses VS via video review also has been a recent focus. The studies to date report reliability for both expert (physicians, neuropsychologists, certified athletic trainers) and non-expert (non-medical) raters [8]. A recent study of the identification of VS of SRC in NHL players by Hutchison and colleagues [8] compared the reliability between non-medical (retired game officials) and expert raters (medical professionals) and reported higher total percent agreement (TPA) and kappa values for the expert compared to the non-expert raters. In contrast, Echemendia et al. [15] enlisted a SRC expert to train a group of non-medical raters to observe and record VS of SRC in concussed NHL players and reported inter-rater agreement ranging from 73% to 99%, which led the researchers to conclude that non-medical personnel can be trained to reliably identify VS of SRC in NHL players.

In the case of American football, using video review to diagnose SRC poses unique challenges. The use of helmets with facemasks can complicate the assessment of certain VS (e.g. blank/vacant look), and helmet adjustments post-impact may be interpreted as the VS of clutching of the head. In addition, multiple head impacts can occur simultaneously among 22 players on any given play. These gameplay features are unique characteristics of American football and the National Football League (NFL). Therefore, the previously published data on the VS of SRC documented in other professional leagues (e.g. NHL, NRL), may not be applicable to NFL players.

The frequency and reliability amongst non-expert and expert raters of VS of SRC in the NFL by video review methodology is understudied. Using data from the 2015 and 2017 NFL seasons, we sought to: (1) assess VS of concussion with the AFL checklist under two conditions (using a single VS in the 2015 season and then utilizing all VS in the 2017 season), (2) document the frequency of VS of SRC; and (3) compare reliability between non-expert and expert raters in the documentation of VS of SRC. Given the equivocal results of the available studies using non-expert and expert raters, we adopted the null hypothesis for the current investigation. It is our hope that by sharing this initial experience of video review at the NFL level, we can gain additional insight into the evolution of video review and how to best assess VS of SRC in professional American football.

Materials and methods

Research design & procedure

A retrospective, observational research design was used for the 2015 season data, which served as the pilot phase of investigation. Based on the results of the pilot study, a prospective, observational research design was used for the second pilot study of the 2017 season data. The current study was approved by the NFL, National Football League Players Association (NFLPA), and an institutional IRB (0006018120). In accordance with the Collective Bargaining Agreement established between the NFL and NFLPA, all athletes consented *a priori* to deidentified injury data use for NFL Health and Safety Initiative research purposes.

Raters

VS data for SRC were recorded by non-expert and expert raters. The previously validated Australian Football League (AFL) Checklist (see Table 1) was utilized by both non-experts and experts for the current studies [5]. Non-expert raters were defined as NFL Player Health and Safety Department employees with a bachelor's education and familiarity with the game of football. Non-expert raters reviewed video footage of 96 cases of SRC from the 2015 season and independently recorded VS of each SRC, selecting a solitary, primary VS from the AFL Checklist. Expert raters were defined as advanced degree biomechanical engineers with significant concussion and head impact research experience who, based on the results of the 2015 pilot study, assessed all relevant VS from the AFL checklist. It must be noted that the clinical concussion experts used in current practice during the 2019–2020 NFL season are physicians and certified athletic trainers, and not the same experts used in the current experimental study [19,20]. Engineers were used in the current study due to their prior involvement in SRC management research and in keeping with prior studies [21]. Expert raters reviewed video footage of 211 cases of SRC from the 2017 season and also independently recorded VSs of SRC for each injury.

Measures and instrumentation

Definition of concussion

An 'injury play' was defined as a single play during which an SRC was determined to occur [22]. This determination was made by the NFL Player Health and Safety Department based on injury data reported by NFL Club physicians and head athletic trainers. Additional video footage were gathered

Table 1. Operational definitions of VS of concussion taken from the Australian Football League (AFL) checklist.

Visible sign	Definition
No Behavior Observed	No observable sign witnessed
Slow to Get Up	Sits or lies on ground despite play continuing; Not held down by another player
Motor Incoordination	Appears clumsy; Unsteady on feet, walk in staggered fashion, or 'rubbery legs'; Unable to pass or receive a ball in a coordinated fashion
Loss of Responsiveness	Lying on ground for more than 1 second; Does not appear to move or react or reply to game situation or others
No Protective Action	No effort to protect himself when falling to the ground; Includes 'rag doll' appearance
Clutching at Head/Face	Reaches for head/face with one or both hands
Blank/Vacant Look	No facial expression or emotion in response to environment or injury
Impact Seizure	Tonic posturing, abnormal sustained muscle contraction; Involves one or both arms; Limb is held stiff despite the influence of gravity
Facial Injury	Facial laceration, bleeding, or eye injury

from the NFL Injury Video Review System, and injury data on SRC diagnosis were obtained from sideline medical evaluations conducted by game-day health-care professionals [19,20,22]. All concussions were diagnosed in accordance with the NFL Game Day Concussion Diagnosis and Management Protocol in place at that time [19,20], and injury details were entered into the NFL's league-wide electronic medical record system.

Video footage of SRC visible signs

NFL Films provided footage from four sources for each injury play. These sources included: (1) wide angle end zone and sideline views, (2) network broadcast coverage, which included all aired footage during the game, (3) untelevised NFL Films footage, often used for documentaries, and (4) network footage not used for televised games. All video footage was edited with in-depth video-editing software, which allowed for zooming and viewing at slowed speeds. (Adobe Premiere Pro; Adobe Systems Incorporated).

SRC visible signs checklist & definitions

All raters were given a pre-specified checklist of VS definitions, which was previously validated in a study of the Australian Football League (AFL) [23]. Definitions of each VS were described in detail for standardization purposes, and the presence or absence of each sign was documented by both the non-expert and expert raters. Each video consisted of the gameplay immediately before the event, for up to approximately 1 min after the event, and slow-motion replay with multiple camera angles. Raters could review the video independently as many times as needed. As mentioned, the AFL checklist was piloted with the 2015 season by non-expert raters, and the same AFL checklist was used prospectively by expert raters for the 2017 season. An important aspect of data collection warrants further emphasis. Since the non-expert raters were engaged in a pilot study during the NFL's first experience with using video review to evaluate signs of SRC, they were asked to record only a single, primary VS. The results of the pilot study led to allowing expert raters to record up to eight VS per injury play for the 2017 season. While these two methods represent the natural evolution of our use of video review, they complicate between-group comparison.

Statistical analysis

The frequency of VS of SRC was summarized for the two non-expert raters using the 2015 dataset and the expert raters using the 2017 dataset. Total percent agreement (TPA) was defined as the total number of times both raters agreed on a VS divided by the total number of injuries and was calculated for each VS in both groups. For example, if one rater recorded a VS in the same 10 of 20 plays, TPA would be 50%. TPA was calculated for each VS in both groups. Despite recording different numbers of VS in the 2015 non-expert and 2017 expert datasets, agreement was calculated between non-experts and experts, who each used the same method, Kappa coefficients (κ) were chosen as the measure of inter-rater agreement due to its use of a correction factor that adjusts for agreement by chance and were calculated for each data set. Kappa coefficients were interpreted with agreement classifications published by Shrout et al. [23] which included: substantial (.81–1.00), moderate (.61–.80), and fair agreement (.41–.60) levels. Statistical significance was set *a priori* at $p < .05$.

Results

Non-expert ratings and reliability of SRC visible signs for the 2015 NFL season

Among the 96 cases of SRC assessed by non-expert raters from the 2015 NFL season pilot study, and utilizing the identification of a single VS, the most frequent VS were *no behavior observed, slow to get up, and motor incoordination*, and the least frequent VS were *impact seizure, blank/vacant look, and facial injury*. The average TPA for the non-expert ratings for each symptom ranged from 84% to 100%. Non-expert raters were incomplete (100%) agreement (i.e. present/not present consensus for the same VS for each video of SRC) for 74% (71/96) of the SRCs. Kappa coefficients for the non-experts ranged from .52. to .68. Kappa coefficients were not calculated for *blank/vacant look, impact seizure*, or *facial injury* due to the absence of these observed signs. There were no VS that attained substantial agreement (.81–1.00) based on the kappa classifications [24]. The non-experts achieved moderate agreement (.61–.80) for 67% (4/6) of VS and fair agreement (.41–.60) for the remaining 33% (2/6) VS (Table 2).

Expert ratings and reliability of SRC visible signs for the 2017 NFL season

Experts reviewed multiple videos associated with 211 cases of SRC from the 2017 NFL season. As mentioned, based on the pilot study of the 2015 data, experts were allowed to record up to 8 VS. The most frequent VS identified from this season were *slow to get up, motor incoordination*, and *no protective action*, and the least frequent VS were *impact seizure, blank/vacant look, and facial injury*. The average TPA for each symptom ranged from 83% to 100%. Expert raters were in complete agreement for 53% (112/211) of the VS associated with SRC video reviews. Kappa coefficients for the experts ranged from .56 to .86. Experts exhibited substantial (.81–1.00) agreement

Table 2. The percentage and total percent agreement (TPA) for VS signs of SRC and kappa coefficients of reliability from two non-expert raters for the 2015 NFL season (n = 96 SRC).

	2015 Season			
On-field sign	Non-expert 1 (%)	Non-expert 2 (%)	TPA (%)	κ
No Behavior	40	47	84	.68*
Slow to Get Up	23	20	89	.66*
Motor Incoordination	18	12	92	.67*
Loss of Responsiveness	9	10	93	.59*
No Protective Action	5	6	95	.52*
Clutching at Head/Face	5	4	97	.65*
Blank/Vacant Look	1	0	99	N/A
Impact Seizure	0	0	100	N/A
Facial Injury	0	0	100	N/A

*$p < .001$.

Table 3. The percentage and total percent agreement (TPA) for VS signs of SRC and kappa coefficients of reliability from two expert raters for the 2017 NFL season SRCs (n = 211).

	2017 Season			
On-field sign	Expert 1 (%)	Expert 2 (%)	TPA (%)	κ
Slow to Get Up	69	63	91	.79*
Motor Incoordination	34	23	83	.59*
No Protective Action	22	13	89	.61*
No Behavior	24	29	91	.78*
Loss of Responsiveness	19	18	96	.86*
Clutching at Head/Face	26	33	84	.63*
Impact Seizure	6	6	97	.73*
Blank/Vacant Look	1	2	99	.56*
Facial Injury	1	0	100	.66*

*p < .001.

Table 4. A comparison of total percent agreement (TPA) and kappa coefficients of reliability for non-expert raters of VS for SRCs from the 2015 NFL season (n = 96) and expert raters of VS for SRCs from the 2017 NFL season (n = 211).

	Non-expert		Expert	
On-field sign	TPA (%)	κ	TPA (%)	κ
Slow to Get Up	89	.66*	91	.79*
Motor Incoordination	92	.67*	83	.59*
No Protective Action	95	.52*	89	.61*
No Behavior	84	.68*	91	.78*
Loss of Responsiveness	93	.59*	96	.86*
Clutching at Head/Face	97	.65*	84	.63*
Impact Seizure	100	N/A	97	.73*
Blank/Vacant Look	99	N/A	99	.56*
Facial Injury	100	N/A	100	.66*

*p < .001.

for 11% (1/9), moderate agreement (.61–.80) for 67% (6/9), and fair agreement (.41–.60) for 22% (2/9) of the SRC VS. The complete frequency, agreement, and reliability data for the 2017 season is presented in Table 3. In addition, a qualitative comparison of the agreement and reliability data between non-expert and expert raters is presented in Table 4.

Discussion

The current study assessed the frequency of VS of SRC for the 2015 and 2017 NFL seasons utilizing varying methodologies and also documented the agreement and reliability of non-expert and expert raters of VS of SRC from these seasons. Overall, certain VS were observed more frequently than others, the TPA for non-expert and experts were different, and reliability values were higher for expert raters compared to non-expert raters. However, different numbers of SRC cases and different rating methodologies were utilized by the expert and non-expert raters, and the preliminary support for the use of expert raters to observe and record VS of SRC in the NFL should be interpreted cautiously. These data may be considered for use by other sports to determine whether individuals with prior experience in that sport rather than those without sport-specific experience can use video review to detect SRC. Overall, these data allow us to gain insight into the complexities of video review and document where we have started and how we have improved methodologically in this realm.

The frequency of the VS of SRC recorded by both non-expert and expert raters across the 2015 and 2017 seasons are comparable to the VS of SRC reported in the NHL and AFL. The most frequent VS reported in the current study were *slow to get up, motor incoordination, no behavior, no protective action*, and *loss of responsiveness*. Alternatively, less frequent VS reported in the current study were *facial injury, blank/vacant look, and impact/seizure*. Facial injuries were nearly absent in the current study but much more prevalent in AFL [14] and NHL [15] studies, which might be explained by the facemask in American football compared to the non-helmeted AFL players and partial facemask used by NHL players. In addition, *blank/vacant look* occurred in 2% of the NFL concussions examined in the current study, which is similar to 1% and 6% reported by the NHL and AFL, respectively [14,15]. Interestingly, another AFL study reported that 75% of SRCs examined involved a *blank/vacant look* VS [24]. The higher percentages reported in this AFL study compared to the current study is likely due to the unobstructed view of the player's face in an un-helmeted, unmasked AFL player. Another plausible explanation for these differences may be due to differences in video quality between professional leagues, as the authors of prior AFL papers reported that detecting a *blank/vacant look* in an injured player was challenging when video quality was limited. Lastly, *impact/seizures* were exceedingly rare in the current study (i.e. 6% of SRCs among expert raters), and in concordance with previous AFL research (i.e. 5%) [14,24]. These findings further highlight the rare occurrence of these VS in concussed athletes.

The agreement and reliability data for VS of SRC reported in the current study are also comparable to previous research. The VS of SRC in NFL players with highest agreement among experts and non-experts were *loss of responsiveness* and *slow-to-get up*, which are consistent with prior AFL studies [14,24]. More specifically, an AFL study reporting head impacts rated by expert and non-expert (referred to as 'naïve') raters revealed that *slow to get up* had the highest intra-class correlation (ICCs = .73-.89) among all eight VS of SRC. The reliability for *loss of responsiveness* was particularly high in the current study among experts (κ = .86), but much lower among non-experts (κ = .66). Inter-rater reliability of *loss of responsiveness* in AFL studies has been fair (ICC = .26–.35), yet intra-rater reliability – reliability within the same rater – was high (ICC = .88 and .70) [14,24]. No obvious explanation exists for this discrepancy, though based on high intra-rater and low inter-rater reliability, we can hypothesize that definitional perceptions among raters may be a factor. Lastly, though *impact seizures* were rare, when they did occur, agreement was moderate among experts (κ = .73).

Several VS of SRC exhibited poor reliability in the current study. The reliability for *blank/vacant look* was fair among experts (κ = .56), which is similar to both NRL [25] and AFL studies [14,24]. An AFL study reported low agreement for *blank/vacant look*, with ICC ranging from .24 to .43, and reliability minimally improved on second review. These lower agreement and kappa values for *blank/vacant look* in the current study may be due to the difficulty for a rater unfamiliar with a player's baseline facial expressiveness to discern a change, and those more familiar with player's baseline facial expressiveness may be more accurate than an unfamiliar rater. Prior studies [15] have reported high agreement among *clutching at head*, whereas the current NFL results were low

(κ = .46 and .63), with non-experts higher than experts, respectively. This likely represents difficulty differentiating between a helmet adjustment and grabbing for the head as a symptom, a nuance of helmeted sports. However, *clutching at head* achieved >70% agreement in 87.6% of NHL concussions, suggesting differences in the lighter, partially-face masked NHL helmet compared to the heavier, masked NFL helmet. Moderate agreement was seen for *motor incoordination* and *no protective action*. Specific to NFL concussions, it is also possible that these VS can be difficult to discern in a crowded huddle, pile of players, special team plays, away-from-ball injuries, or multi-player collision(s).

Despite variability in a number of videos reviewed and VS determination methods, raters exhibited greater agreement for identifying VS of SRC compared to non-experts. Experts achieved substantial agreement for kappa coefficient values across one VS, and non-experts did not reach substantial agreement on any VS. Further, expert raters achieved moderate agreement levels for kappa across six VS, whereas non-experts achieved a similar level of agreement on four. As stated, non-experts recorded a single VS, while experts recorded multiple VS, which is an acknowledged limitation of this study, but reflects the evolution of the video review process in the NFL over time. The evidence supporting expert or non-expert raters for identifying VS of SRC is mixed across professional sports. The superiority of experts was demonstrated in the NRL [25], where kappa values were higher among experts (those with sideline experience managing SRC) compared to novice raters (limited experience with NRL play and no prior SRC management). Conversely, a similar study of NHL concussions found no difference in kappa values between non-SRC specialists evaluating SRC [15]. However, the NHL raters had substantial hockey knowledge and were trained by concussion specialists until agreement reached 80% or better. Future studies might assess the utility of training non-expert raters to evaluate VS in NFL SRC, especially given the complexities and nuance of surveying a large field with 22 players.

Though the current study is limited by varying data collection methods, we can take a step back and see how video review has evolved at the NFL level. One decade ago, video review was not used routinely in the evaluation of SRC. Fast forward to today's game, the current NFL Concussion Diagnosis and Management Protocol is a complex, thorough, multi-faceted system, with video review well integrated and mandated [26]. In today's clinical gameday practice, the NFL enlists a minimum of five health-care professionals per game to monitor for concussions. Two unaffiliated neurotrauma consultants (UNCs) – physicians with experience treating brain injuries – are on each sideline, and one is in an above field box monitoring the game. Also in the box are two certified athletic trainers (ATC) – referred to as spotters – that help identify events suspicious for concussion using video feed [19,20]. While the two UNCs on the field are available for immediate SRC assessment, the single UNC and two ATC spotters are in the private box and are viewing live, broadcast feed video. All evaluators in the private box can stop play at any time based on the video feed, and when needed, a sideline video monitor is made available to the two on-field UNCs [18]. Using the SRC game-day checklist, four 'No-Go' signs – LOC, confusion, amnesia, and impact seizure – call for immediate removal, and broadcast video is used to identify any one of these signs by any of the five game-day experts. The current pilot, laboratory studies are intended to enhance the ability of gameday clinicians to identify VS of concussions in NFL players. The preliminary data from our studies provided a platform from which to improve the identification of VS in SRC, and it is our hope that researchers in other sports can use this methodology to similarly improve SRC detection.

Limitations and future research

There are several limitations to the current study, which further underscore the preliminary, hypothesis-generating nature to these findings. Most importantly, there was inconsistency of data collection methods and cases assessed between non-expert and expert raters. The non-experts (for the 2015 data) utilized the hierarchical flowchart described by Davis and Makdissi [5] in which only the highest ranked sign was recorded. Thus, only one sign was recorded for each case. As mentioned, our purpose in utilizing non-experts for the 2015 data was to perform a pilot study on the feasibility of using the AFL VS checklist for NFL concussions. The lessons learned from the 2015 pilot study led the experts to record the presence or absence of all eight signs for all of the 211 cases across the 2017 season data. Since the non-experts did not record all signs and were limited to just the single highest ranked sign, their agreement may appear artificially lower than it would be if all signs had been recorded. Additionally, non-expert and expert raters did not review the same videos, and non-experts reviewed fewer videos than experts, which limits the direct comparison and statistical testing of reliability between these groups. Despite these methodological limitations, we view the current results as useful in describing laboratory investigation of VS of concussion and for informing future studies of VS of SRC in the NFL. The sample sizes used in the current study were variable, which may limit the generalizability of these findings to non-expert and expert raters, as well as to nonprofessional American football players. Lastly, as described, the expert raters used in the current study are not the same medical experts used during gamedays in the NFL. Current practice uses team physicians and athletic trainers to monitor video review [19,20], and though PhD level engineers are not used in game, they have been used in prior laboratory studies for evaluation of injury behaviors and biomechanics [22]. Future research should directly compare the frequency, agreement, and reliability of non-expert and expert raters of VS reviewing the same SRC in NFL players. Finally, the publication of international consensus visible signs of concussion suggests that this assessment methodology may yield better reliability values going forward [18,27].

Conclusions

In the current preliminary analysis of VS for SRC in NFL players, frequencies of several VS were documented across two seasons in two different rater groups. Despite varying methodologies, the preliminary results indicated that the reliability of VS may be stronger for experts compared to non-experts due to gameplay nuances and protective equipment used in the NFL compared to other sports (e.g. hockey and rugby). It is our

hope these early studies of video review of SRC at the NFL level can improve the state of video review of SRC.

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Declaration of interest

No potential conflict of interest was reported by the authors.

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